Investment Evaluation Exercise – Company GREENPACK

GREENPACK operates in the sustainable packaging industry. Its flagship product is the ECOBOX container, which is sold at a price of €120 per unit, with avoidable (variable) production costs equal to €65 per unit.

In 2025, management is evaluating a new investment opportunity. GREENPACK could purchase a new automated production line to manufacture a next-generation biodegradable container called BIOPACK, which would be sold at €160 per unit, with avoidable production costs equal to €80 per unit.

The introduction of BIOPACK is expected to reduce demand for ECOBOX.

The expected sales forecasts for the four-year period 2027–2030 are shown below.

Year	2027	2028	2029	2030
Demand BIOPACK (units)	8,000	9,000	10,000	11,000
Reduction Demand ECOBOX (units)	5,000	5,500	6,000	6,500

To produce BIOPACK, GREENPACK must purchase a new automated production line at a cost of €1,000,000.

The production line:

- is purchased on Dec 31, 2026
- becomes immediately operational
- has a fiscal depreciation life of 6 years
- is depreciated using the straight-line method

The project lasts 4 years. At the end of 2030 the production line is sold for €250,000. When evaluating the investment, the following additional effects must also be considered:

Two technicians must be hired. Annual salary per technician: €40,000

Training costs will be incurred as follows:

Year	Training Cost
2027	€40,000
2028	€30,000
2029	€0

Year	Training Cost
2030	€0

Additional administrative costs

Year	Administrative Cost
2027	€20,000
2028	€25,000
2029	€30,000
2030	€30,000

Financing

The company obtains a subsidized government loan of: €250,000 at an annual interest rate of 4% starting January 1, 2027. For simplicity, assume that the debt remains outstanding throughout the project life and is refinanced indefinitely. Therefore, no debt repayment occurs during the forecast horizon.

Assume the following:

- Corporate tax rate: 30%
- Cost of equity: 22%
- No changes in working capital

Questions

1. Compute the Free Cash Flow to the Firm (FCFF) generated by the project.
2. Compute the project's Net Present Value (NPV) using the FCFF approach.